

Hemoglobinopathies & Thalassemias - Factoid Sheet

Week 7 Hematology • Raval • memorize-first companion

ScholarRx: [Hemoglobin Structure & Function](#) • [Sickle Cell Disease](#) • [Genetic Basis of Select Thalassemias](#)

Core frame: structural/qualitative = wrong globin sequence; thalassemia/quantitative = wrong amount of globin. Ask every composition question: **Is HbA present? Is HbA2 high? Is HbF high? What variant replaces A?**

Normal Hb - memorize cold

Hb	Chains	Adult
A	alpha2 beta2	96-98%
A2	alpha2 delta2	2.5-3.5%
F	alpha2 gamma2	<1%

- **alpha cluster:** chr16; 4 alpha genes total.
- **beta cluster:** chr11; 2 beta genes total.
- HbF higher O2 affinity; inhibits HbS polymerization.

Qualitative vs quantitative

HbS/C/E: structural variants, usually missense.

Thal: decreased/absent chain synthesis.

- Alpha-thal: usually deletions.
- Beta-thal: promoter/splice/nonsense/frameshift etc.
- **beta+** reduced; **beta0** absent.

Sickle cell - mutation to complication

Fact	Must know
Mutation	beta6 Glu -> Val; GAG -> GTG
Mechanism	Deoxy HbS polymerizes -> rigid RBCs -> vaso-occlusion + membrane damage/hemolysis.
Promotes sickling	Hypoxia, acidosis, dehydration / high MCHC.
Protective	HbF (and HbA) inhibits polymerization.
Vaso-occlusive clues	Pain, dactylitis, ACS, stroke, AVN, priapism, nephropathy.
Hemolysis clues	Chronic anemia, retics, bilirubin/gallstones, NO scavenging/endothelial dysfunction.
Classic associations	Functional asplenia -> encapsulated infection; Salmonella osteomyelitis; parvovirus B19 aplastic crisis.

Structural variants

HbS: beta6 Glu->Val; polymerizes.

HbC: beta6 Glu->Lys; mild hemolysis/splenomegaly; crystals.

HbE: beta26 Glu->Lys; Southeast Asia; structural variant + mild beta-thal effect via altered splicing.

Hb analysis fast rules

- **AS trait:** A > S.
- **SS:** no A; S dominant.
- **SC:** S + C; no A.
- **beta-thal trait:** A dominant + **high A2**.
- **beta-thal major:** high F, very low/absent A.
- **alpha-thal trait:** adult electrophoresis may look normal.

Thalassemia - gene count, chain imbalance, phenotype

Alpha-thal deletion ladder

Deleted	State	Signature
1	Silent carrier	Usually minimal findings
2	Trait	Cis (--/aa) or trans (-a/-a); microcytosis
3	HbH disease	beta4 HbH ; hemolytic anemia
4	Hydrops fetalis	gamma4 Hb Barts ; extreme O2 affinity / severe fetal disease

Alpha: delete genes and count what is left. **Beta:** ask beta+ or beta0.

Beta-thal phenotype

Form	Key findings
Trait/minor	Heterozygous; mild/asymptomatic microcytic anemia; HbA2 elevated (~5% on slide) ; mild HbF increase possible.
Intermedia	Moderate anemia; variable transfusion; hepatosplenomegaly, skeletal changes, extramedullary hematopoiesis.
Major	Severe early transfusion dependence; excess alpha chains -> ineffective erythropoiesis + hemolysis; marrow expansion; growth/bone changes; iron overload.

Why beta-thal hurts marrow

Unpaired **alpha chains are insoluble** -> precipitate in erythroid precursors -> apoptosis/ineffective erythropoiesis. Severe anemia -> EPO -> marrow expansion + extramedullary hematopoiesis.

Iron overload

Comes from **transfusions + increased intestinal absorption** driven by ineffective erythropoiesis/hepcidin suppression.

Treatment factoids + 2026 corrections

Therapy	Factoid
Hydroxyurea	Raises HbF -> anti-sickling.
Transfusion	Severe anemia / stroke prevention; creates iron burden.
Chelation	Needed for chronic transfusional iron overload.
HSCT	Curative-intent for selected SCD/thal patients.
Casgevy	Genome editing strategy -> increase HbF; current FDA indications include recurrent-VOC SCD and transfusion-dependent beta-thalassemia.
Lyfgenia	Anti-sickling beta-globin gene addition for SCD.
Zynteglo	Gene-addition therapy for transfusion-dependent beta-thalassemia.
Mitapivat (Aqvemes)	FDA 2026: oral PK activator for anemia in adults with alpha- or beta-thalassemia.
Voxelotor	Do not memorize as current therapy: voluntarily withdrawn from market in 2024 after FDA safety alert.

If you see X -> think Y

- Microcytosis + high HbA2 -> **beta-thal trait**.
- Newborn initially well, later SCD symptoms as HbF falls -> **gamma-to-beta switch**.
- S + C with no A -> **HbSC disease**.
- Four alpha deletions + fetal hydrops -> **Hb Barts (gamma4)**.
- Three alpha deletions -> **HbH (beta4)**.
- Severe anemia + maxillary/skull expansion -> **ineffective erythropoiesis / marrow expansion**.
- CRISPR target BCL11A enhancer -> **reactivate gamma globin / HbF**, not direct HbS correction.

Last 5 minutes: A=alpha2beta2, A2=alpha2delta2, F=alpha2gamma2. S=Glu6Val, C=Glu6Lys, E=Glu26Lys. Alpha deletion count: 1 silent, 2 trait, 3 HbH beta4, 4 Barts gamma4. Beta trait = high A2. HbF protects SCD and beta-thal.

Primary sources: Raval/Kutlar lecture deck + uploaded Lecture 9 companion. Current therapy notes are explicitly external FDA updates checked in 2026.